

CSES — String Algorithms Problem Reference

All tasks: 1 s, 512 MiB. Counts are modulo $10^9 + 7$ where stated.

1731 — Word Combinations

Given lowercase string s and a dictionary of k distinct words, count the ordered ways to split all of s into dictionary words (words may be reused). **In:** s ; k ; then k words

Out: number of complete segmentations mod $10^9 + 7$

Constraints: $|s| \leq 5000$, $k \leq 10^5$; total dictionary length $\leq 10^6$.

1753 — String Matching

Given lowercase text s and pattern p , count starting positions where p occurs in s ; overlapping occurrences count. **In:** s ; p

Out: number of occurrences

Constraints: $1 \leq |s|, |p| \leq 10^6$.

1732 — Finding Borders

A border is a proper prefix that is also a suffix. Find the lengths of all borders of a lowercase string. **In:** s

Out: all border lengths in increasing order

Constraints: $1 \leq |s| \leq 10^6$.

1733 — Finding Periods

A period is a prefix whose repetitions generate the whole string; the final repetition may be partial. Find all valid period lengths (including $|s|$). **In:** s

Out: all period lengths in increasing order

Constraints: $1 \leq |s| \leq 10^6$.

1110 — Minimal Rotation

Among all cyclic rotations of a lowercase string, output the lexicographically smallest rotation. **In:** s

Out: lexicographically minimal rotation

Constraints: $1 \leq |s| \leq 10^6$.

1111 — Longest Palindrome

Find a longest palindromic *substring* of a lowercase string. If several exist, output any one. **In:** s

Out: any longest palindromic substring

Constraints: $1 \leq |s| \leq 10^6$.

3138 — All Palindromes

For every position i , find the maximum length of a palindromic substring that ends exactly at i . **In:** s

Out: n lengths, for positions $1, \dots, n$

Constraints: $1 \leq n \leq 2 \cdot 10^5$.

1112 — Required Substring

Count uppercase A–Z strings of length n that contain a given pattern p as a contiguous substring at least once. **In:** n ; pattern p

Out: count mod $10^9 + 7$

Constraints: $1 \leq n \leq 1000$, $1 \leq |p| \leq 100$.

2420 — Palindrome Queries

Maintain a lowercase string under: 1 **k** **x** set $s_k = x$; 2 **a** **b** ask whether $s[a..b]$ is a palindrome. **In:** n, m ; s ; then m operations

Out: YES/NO for each type 2 query

Constraints: $n, m \leq 2 \cdot 10^5$.

2102 — Finding Patterns

Given a lowercase text and many lowercase patterns, decide independently whether each pattern occurs as a substring. **In:** s ; k ; then k patterns

Out: YES or NO for each pattern

Constraints: $|s| \leq 10^5$, $k \leq 5 \cdot 10^5$; total pattern length $\leq 5 \cdot 10^5$.

2103 — Counting Patterns

For each of many patterns, count its starting positions in a fixed text; overlapping occurrences count. **In:** s ; k ; then k patterns

Out: occurrence count for each pattern

Constraints: $|s| \leq 10^5$, $k \leq 5 \cdot 10^5$; total pattern length $\leq 5 \cdot 10^5$.

2104 — Pattern Positions

For each pattern, find the first 1-indexed position where it occurs in a fixed text, or -1 if absent. **In:** s ; k ; then k patterns

Out: first position or -1 for each pattern

Constraints: $|s| \leq 10^5$, $k \leq 5 \cdot 10^5$; total pattern length $\leq 5 \cdot 10^5$.

2105 — Distinct Substrings

Count the distinct nonempty substrings of a lowercase string. **In:** s

Out: number of distinct substrings

Constraints: $1 \leq |s| \leq 10^5$; answer fits 64-bit.

1149 — Distinct Subsequences

Delete arbitrary characters without reordering the rest. Count the distinct *nonempty* strings obtainable as subsequences. **In:** s

Out: number of distinct nonempty subsequences mod $10^9 + 7$

Constraints: $1 \leq |s| \leq 5 \cdot 10^5$.

2106 — Repeating Substring

Find a longest substring that occurs in at least two positions of s (occurrences may overlap). If none exists, print -1 . **In:** s

Out: any longest repeating substring, or -1

Constraints: $1 \leq |s| \leq 10^5$.

2107 — String Functions

For every 1-indexed position i , compute $z(i)$ = longest prefix match starting at i (with $z(1) = 0$), and $\pi(i)$ = longest proper prefix ending at i . **In:** s

Out: one line of all z values; one line of all π values

Constraints: $1 \leq |s| \leq 10^6$.

3225 — Inverse Suffix Array

Given a permutation of $1, \dots, n$ claimed to be the lexicographic order of suffix starting positions, construct any lowercase a–z string having exactly that suffix array. **In:** n ; suffix-array permutation

Out: any valid lowercase string; -1 if impossible

Constraints: $1 \leq n \leq 10^5$.

1113 — String Transform

Invert the Burrows–Wheeler transform defined by appending unique sentinel $\#$ (smaller than a–z), sorting all rotations, then taking their last characters. **In:** transformed string of length $n + 1$, containing one $\#$

Out: original lowercase string of length n

Constraints: $1 \leq n \leq 10^6$.

2108 — Substring Order I

Sort all *distinct* nonempty substrings lexicographically and output the k th one. **In:** s ; k

Out: k th distinct substring

Constraints: $|s| \leq 10^5$, $1 \leq k \leq n(n + 1)/2$; k is guaranteed valid.

2109 — Substring Order II

Sort all substring *occurrences* lexicographically (duplicates appear with multiplicity) and output the k th one. **In:** s ; k

Out: k th substring in the multiset order

Constraints: $|s| \leq 10^5$, $1 \leq k \leq n(n + 1)/2$.

2110 — Substring Distribution

For each length $L = 1, \dots, n$, count how many distinct substrings of s have length exactly L . **In:** s

Out: n counts for lengths $1, \dots, n$

Constraints: $1 \leq |s| \leq 10^5$.